



**DEPARTAMENTO DE ELETRÓNICA, TELECOMUNICAÇÕES
E INFORMÁTICA**

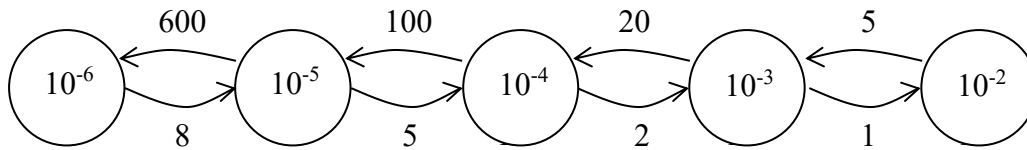
LICENCIATURA EM ENGENHARIA DE COMPUTADORES E INFORMÁTICA

ANO 2025/2026

**MÉTODOS PROBABILÍSTICOS PARA ENGENHARIA DE
COMPUTADORES E INFORMÁTICA**

PRACTICAL GUIDE NO. 6

1. Consider a wireless link between multiple stations for data communications. Consider the bit error rate (*ber*) introduced by the wireless link (due to the variation of the propagation and interference factors along with time) given by the following Markov chain:



where the state transition rates are in number of transitions per hour. Assume that all stations detect with a probability of 100% when the data frames sent by the other stations are received with errors.

- (a) Determine the probability of the link being on each state.
 - (b) Determine the average time percentage the link is on each state.
 - (c) Determine the average holding time (in minutes) of the link on each state.
 - (d) Determine the probability of a data frame of 100 Bytes to be received with errors (i.e., with at least one bit in error).
 - (e) Determine the probability of a data frame of 1500 Bytes to be received without errors.
 - (f) When a data frame of 500 Bytes is received with errors, what is the probability of the link being in state 10^{-6} ? And in state 10^{-2} ? Compare these probabilities with the probabilities in **1(a)** and draw conclusions.
2. Consider a company providing all services on a single server. The server can break down due to 2 causes: hardware failure or software failure. The amount of time the server runs until a hardware/software failure is exponentially distributed with an average of 500 days and 90 days, respectively. The hardware/software repair time is exponentially distributed with an average of 1 day and 4 hours, respectively. A hardware failure requires first the hardware repair and then the software repair, while a software failure requires only the software repair.
 - (a) Draw the state transition diagram of the Markov chain modeling this system with state transition rates in number of transitions per day.
 - (b) Compute the limiting probabilities of each state.
 - (c) What is the probability (in %) of the server being available?
 - (d) What is average time (in hours) that the server is down in one month of 30 days?
 3. Consider a Data Center with 100 servers and t maintenance technicians. The amount of time each server runs until it fails is exponentially distributed with an average of 200 days. The server repair time of each technician is exponentially distributed with an average of 36 hours. Consider a birth-dead process with states $n = 0, 1, \dots, 100$ representing the number of failing servers.
 - (a) Consider $t = 1$ technician. Compute first the state limiting probabilities and then:
 - i) determine the average number of servers down,
 - ii) determine the average number of technicians in repairing activities,
 - iii) determine the probability of at least 90 servers being in operation.
 - (b) Repeat **3(a)** considering $t = 2$ technicians.
 - (c) Repeat **3(a)** considering $t = 3$ technicians. Compare all results and draw conclusions.

4. Consider a $M/M/1$ queuing system such that the arriving rate of clients is $\lambda = 5$ clients per second and the average serving time of each client is $1/\mu = 0.1$ seconds.
 - (a) What is the average time each client stays in the system?
 - (b) What is the average server occupation (in %)?
 - (c) What is the average queue occupation, i.e., the average number of clients in the queue?
 - (d) What is the probability (in %) of the queue occupation becoming larger than 6 clients? And larger than 10 clients? And larger than 20 clients?
 - (e) Using Matlab *plot* function, plot a graph of the average time each client stays in the system as a function of the client arriving rate λ (from $\lambda = 1$ to $\lambda = 9$).

5. Consider a $M/M/1/m$ queuing system such that the arriving rate of clients is $\lambda = 80$ clients per second, the average serving time of each client is $1/\mu = 0.01$ seconds and the system capacity is $m = 10$ clients. Clients are discarded when they arrive and the system is full.
 - (a) What is the probability of each client being discarded?
 - (b) What is the average time each client stays in the system when it is not discarded?
 - (c) What is the average queue occupation?
 - (d) Using Matlab *bar* function, plot a bar chart of the client discard probability as a function of the system capacity m (for $m = 5, 10, 15, \dots, 50$).